

Mahmoud Elsharawy

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EDUCATION

Cornell University

Masters of Engineering in Computer Science

Bachelors of Science in Computer Science & Electrical and Computer Engineering

GPA: 3.78

May 2025

Dec 2024

EXPERIENCE

Apple

System Electrical Engineer

July 2025 - Present

Cupertino, CA

- Integrated schematics for PCBA development, coordinating with Mechanical Engineers, Design for Manufacturing, and PCB Designers to create production-worthy PCBAs
- Characterized and debugged hardware using oscilloscopes and signal analyzers to ensure reliability in production

Apple

System Electrical Engineering Intern

May 2024 - Aug 2024

Cupertino, CA

- Developed a Python tool to automate HSPICE simulations for existing Cadence designs and GPIO specifications

Apple

System Electrical Engineering Intern

Jan 2023 - Aug 2023

Cupertino, CA

- Wrote firmware in C for an SAMD21 microcontroller on a custom PCBA to calibrate its ADC and control a precise micro-current load to characterize power components
- Designed and tested a buck converter power module for use on internal dev boards, removing reliance on a vendor's power modules, preventing future supply chain issues and reducing cost

SpaceX

Hardware Engineering Intern

Jan 2022 - Aug 2022

Hawthorne, CA

- Created a Python package to use SCPI commands and REST API interaction to collect and model hardware data
- Automated an assembly line station through mechanical design and PLC TwinCAT software, tripling its speed
- Tested and qualified alternative integrated circuits for Business User Terminals, preventing a parts shortage

PROJECTS

RP2040 Audio Communication | C++, Embedded Systems, Microcontrollers

Mar 2024 - April 2024

- Developed firmware for a 2-way audio communication protocol on Raspberry Pi RP2040 boards using frequency-based bit encoding with Direct Digital Synthesis, enabling arbitrary ASCII character data transmission via sound
- Implemented automatic threshold detection and PLL synchronization logic for each board to detect when the other board starts transmitting and adjust its clock accordingly

Discord Hard Drive | C++, REST API, Linux, Git

Feb 2025 - Mar 2025

- Developed a Discord bot in C++ that interfaces with Discord's REST API to store arbitrary data through reading, writing, and modifying messages
- Created an nbdkit plugin in C++ to integrate the bot with a virtual block device, enabling seamless read/write operations and allowing Discord messages to be used as a hard drive

Chess AI | Rust, Python, REST API, Systemd, Git

Aug 2021 - Present

- Developed a UCI Chess AI in Rust using a minimax algorithm with move-ordering and variable depth control
- Implemented Zobrist hashing with a hash-table to look up previously seen positions, reducing computation time
- Hosted a systemd service to interact with lichess.com's REST API, allowing the AI to play against other bots and humans, earning an Elo rating of 1850, making it stronger than 85% of human players on the website

TECHNICAL SKILLS

Languages: C/C++, Rust, Java, Python, ARM, RISC-V

Developer Tools: CMake, Git, Linux, Nix/NixOS, Cargo